



SCHOOL OF COMPUTER TECHNOLOGY

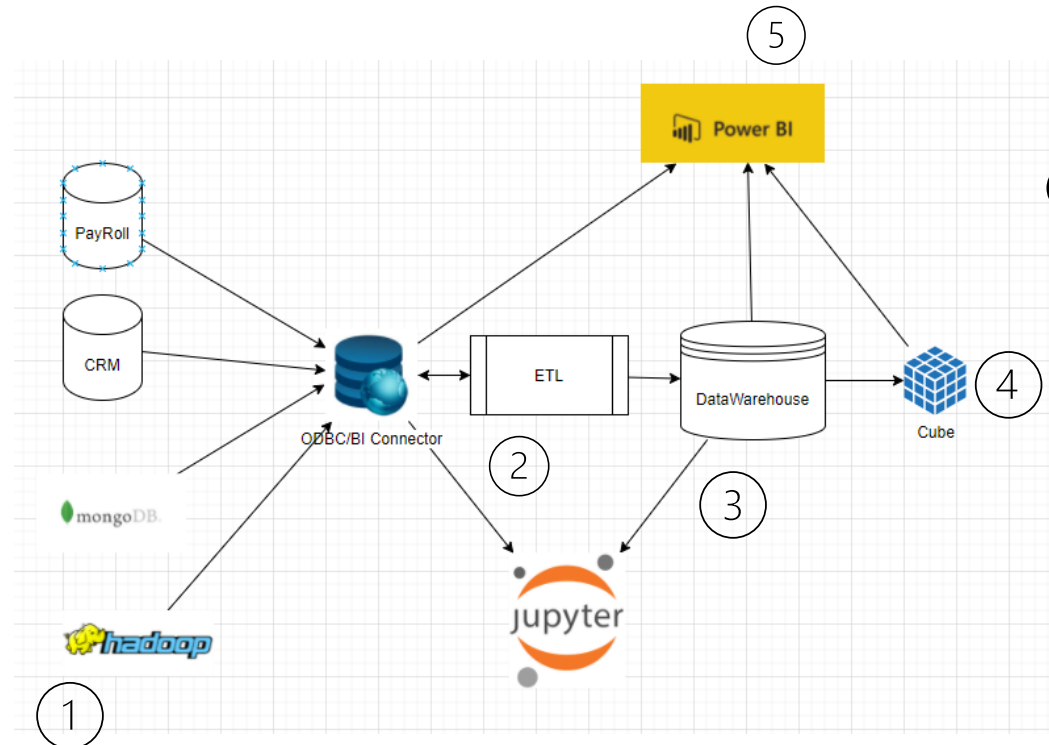
AASD 4012 Big Data Tools and Technologies

High Level Design of a Datawarehouse system

Let's split the high-level design into 5 Steps

1. Operational data source
2. ETL
3. Data Warehouse
4. Cube
5. Data visualization

Read data from operational sources, apply lookups, transformations and aggregations and deposit that data into a data warehouse, further if required, apply a cube on top of the data warehouse and apply data visualization tools on top of databases and data warehouses



High Level Design of a Datawarehouse system – Continued!



Operational Sources	ETL	Cube	Datawarehouse	Data Visualization
CSV/JSON/Text files	Hive /Pig /Spark	Multidimensional	HIVE	PowerBI
Structured data (SQL Server/ My SQL /Oracle...)	SSIS	Tabular	Azure Synapse	Python
Online Feeds (Facebook/ Twitter..)	TSQL		Teradata	Tableau
HDFS	Talend / Pentaho		SQL Server	Power Pivot / Excel
Unstructured data (Emails , notes, comments)	Oracle Data designer		Azure SQL	SSRS
	Informatica			
	Ateryx			



Please also refer to references slide at the end of this presentation for credits and citations for the contents that are not exactly copied but taken as a knowledge or reference

Multidimensional Schema

Multidimensional Schema is specially designed to model data warehouse systems. The schemas are designed to address the unique needs of huge databases designed for analytical purpose (OLAP).



Following are three main types of multidimensional schemas, each having its unique advantages.

- Star Schema
- Snowflake Schema
- Galaxy Schema

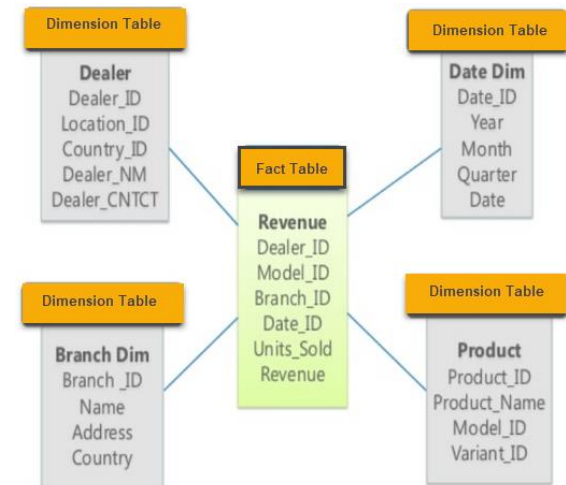


Star Schema

In the data warehouse, in which the star center can have one fact table and several associated dimension tables. It is known as star schema as its structure resembles a star.

The Star Schema data model is the simplest type of Data Warehouse schema. It is also known as Star Join Schema and is optimized for querying large data sets.

In the following Star Schema example, the fact table is at the center, which contains keys to every dimension table like Dealer_ID, Model ID, Date_ID, Product_ID, Branch_ID & other attributes like Units sold and revenue.



Star Schema - Continued



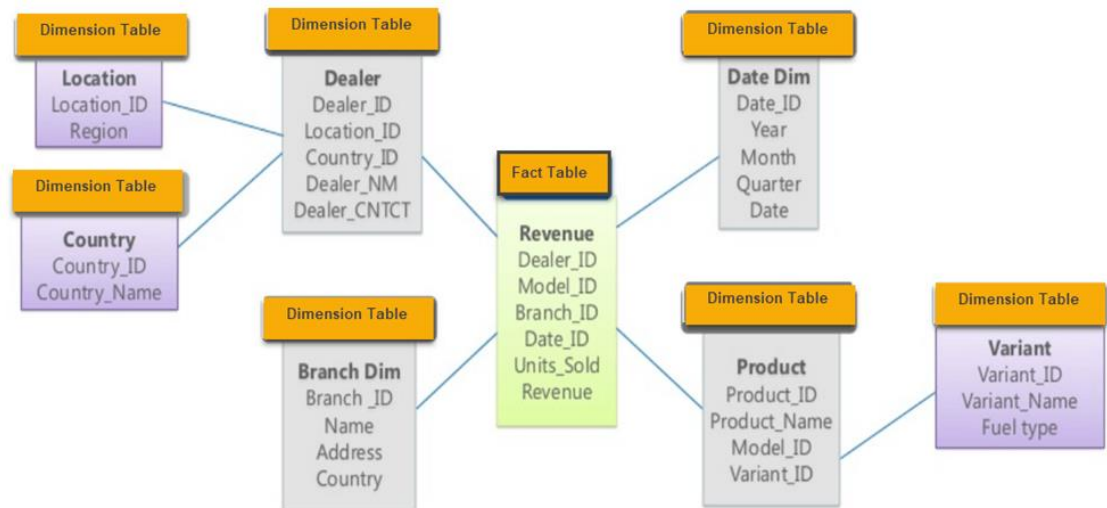
Characteristics of Star Schema:

- Every dimension in a star schema is represented with only one dimension table.
- The dimension table should contain the set of attributes.
- The dimension table is joined to the fact table using a foreign key.
- The dimension table are not joined to each other.
- Fact table would contain key and measure.
- The Star schema is easy to understand and provides optimal disk usage.
- The dimension tables are not normalized. For instance, in the above figure, Country_ID does not have a Country lookup table as an OLTP design would have.
- BI Tools widely support the schema.



Snowflake Schema in the data warehouse is a logical arrangement of tables in a multidimensional database such that the [ER diagram](#) resembles a snowflake shape. A Snowflake Schema is an extension of a Star Schema, and it adds additional dimensions. The dimension tables are normalized, which splits data into additional tables.

In the following Snowflake Schema example, the Country table is further normalized into an individual table. ★



Snowflake - Continued



Characteristics of Snowflake Schema:

- The main benefit of the snowflake schema it uses smaller disk space.
- Easier to implement a dimension is added to the Schema.
- Due to multiple tables query performance is reduced.
- The primary challenge that you will face while using the snowflake Schema is that you need to perform more maintenance efforts because of the more lookup tables.

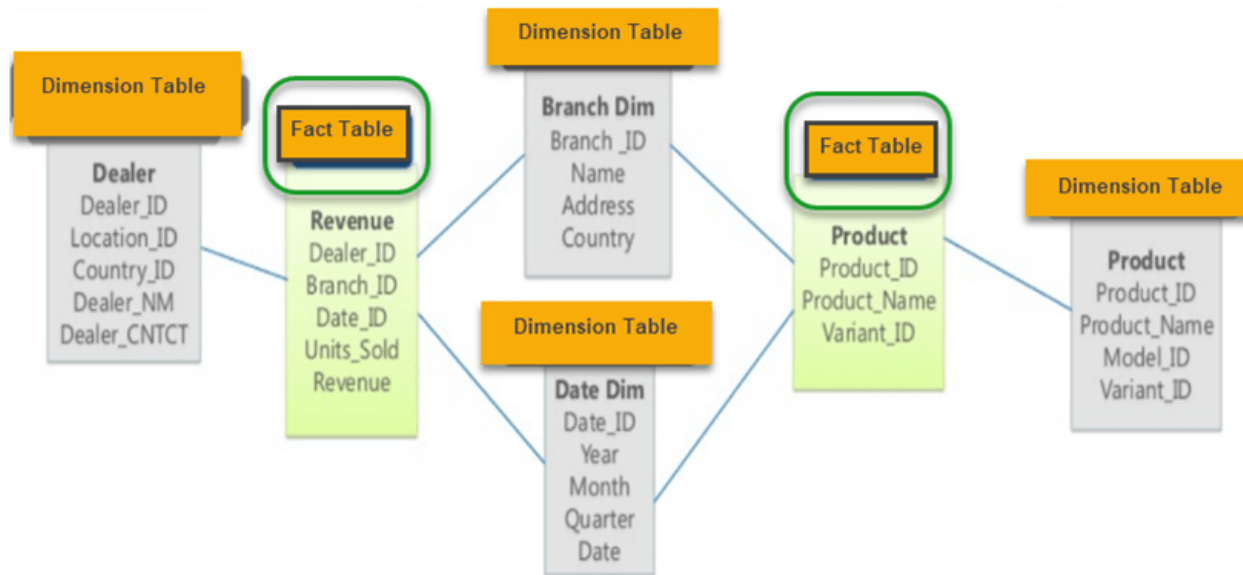


Galaxy Schema

A Galaxy Schema contains two fact table that share dimension tables between them. It is also called Fact Constellation Schema. The schema is viewed as a collection of stars hence the name Galaxy Schema.

As you can see in given example, there are two facts table. **Revenue** and **Product**.

In Galaxy schema shares dimensions are called **Conformed Dimensions**.



Galaxy - Continued



Characteristics of Galaxy Schema:

- The dimensions in this schema are separated into separate dimensions based on the various levels of hierarchy.
- For example, if geography has four levels of hierarchy like region, country, state, and city then Galaxy schema should have four dimensions.
- This schema is helpful for aggregating fact tables for better understanding.
- Large enterprise Datawarehouse are build upon Galaxy schema



Key differences between Star Schema and Snowflake

Star Schema	Snowflake
Hierarchies for the dimensions are stored in the dimensional table.	Hierarchies are divided into separate tables.
It contains a fact table surrounded by dimension tables.	One fact table surrounded by dimension table which are in turn surrounded by dimension tables
In a star schema, only single join creates the relationship between the fact table and any dimension tables.	A snowflake schema requires many joins to fetch the data.
Simple DB Design.	Very Complex DB Design.
Denormalized Data structure and query also run faster	Normalized Data Structure.
High level of Data redundancy	Very low-level data redundancy
Single Dimension table contains aggregated data.	Data Split into different Dimension Tables.
Cube processing is faster.	Cube processing might be slow because of the complex join.



Adventure works 2022 DW



A close to perfect design

Anomalies

- DimProductSubCategory and DimProductCategory are in snowflake . Must have a good reason
- DimGeography is in snowflakes but it has a good reason ?
 - Joined by 2 different dim tables ?
- DimDate is connected to Fac Table by 3 keys
 - Orderdate
 - Duedate
 - Shipdate



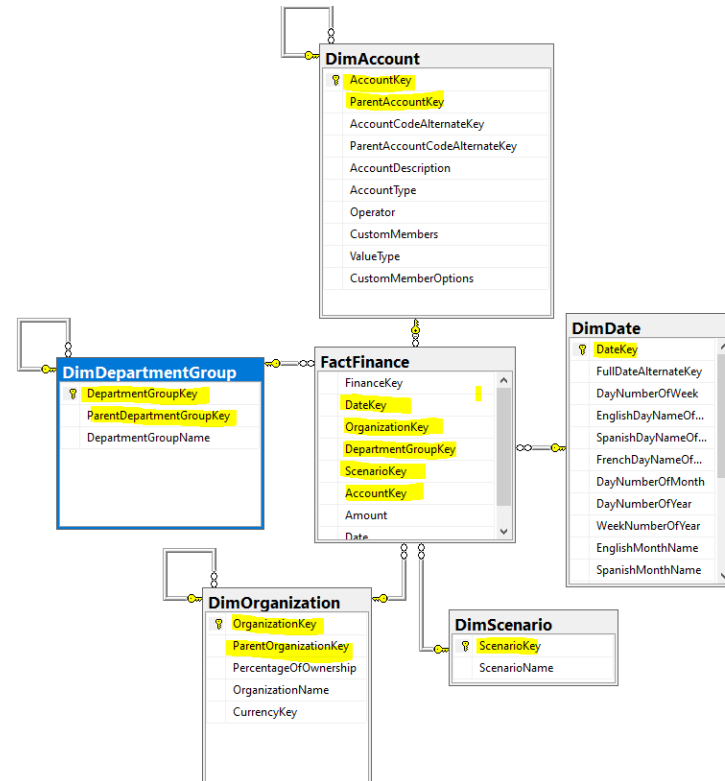
Adventure works 2022 DW



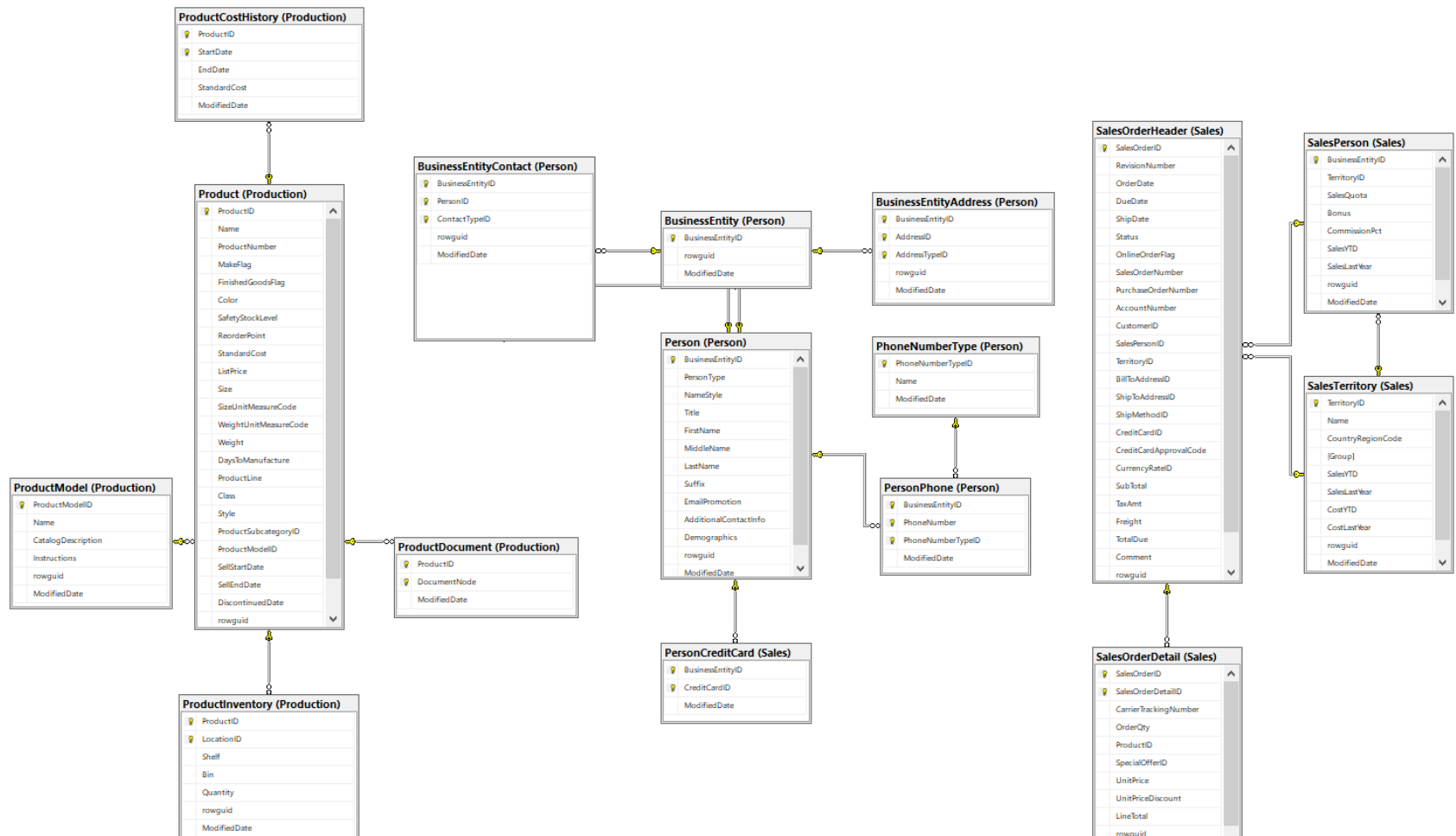
A close to perfect design

New knowledge

- DimAccount, DimDepartmentGroup and DimOrganization are in hierarchy and self join based on parentkey



Adventure works 2022 Operational



ACID - Concept



ACID in databases refers to a set of principles that ensure reliable and consistent transactions.

Atomicity means that a transaction is treated as a single unit of work. Either all the changes within a transaction are applied, or none of them are.

Consistency ensures that a transaction brings the database from one valid state to another, adhering to predefined rules and constraints.

Isolation guarantees that concurrent transactions do not interfere with each other. Each transaction operates independently until it is completed.

Durability ensures that once a transaction is committed, its changes are permanently stored and will survive system failures.

In simple terms, ACID ensures that database transactions are reliable, consistent, isolated, and durable.



ACID - Example



Imagine you have a banking application where users can transfer money between their accounts. Here's how ACID principles apply:

Atomicity ensures that a money transfer transaction is treated as a single unit. If a user transfers \$100 from Account A to Account B, either the entire transfer is completed, and \$100 is deducted from Account A and added to Account B, or nothing happens at all. There won't be a scenario where only \$50 is deducted from Account A, leaving the transaction incomplete.



Consistency guarantees that the database remains in a valid state throughout the transaction. For example, before the transfer, both Account A and Account B have sufficient funds. After the transfer, the total balance in the system should remain the same, and neither account should have a negative balance. Consistency ensures that predefined rules, such as account balances being non-negative, are maintained.

Isolation ensures that concurrent transactions don't interfere with each other. Let's say two users initiate separate transfers simultaneously, one from Account A to Account B and another from Account B to Account C. Isolation guarantees that these transactions will not impact each other's data. Each transaction will operate independently, ensuring the correct transfer of funds between the accounts involved.

Durability: Durability ensures that once a transaction is committed, it persists even in the face of system failures. Suppose the money transfer is successfully completed and committed. Even if a power outage or system crash occurs immediately after the transaction, the transferred amount will still be reflected in the database when the system is restored. Durability ensures the permanent storage and availability of committed transactions.



Please also refer to references slide at the end of this presentation for credits and citations for the contents that are not exactly copied but taken as a knowledge or reference

Characteristics – Operational - DW



An **operational database** is designed to support day-to-day operations of an organization. It is optimized for **quick write and quick read** operations. In operational databases, the primary focus is on efficiently capturing and retrieving data related to **current transactions or real-time activities**. These databases are typically used to manage and process operational data such as customer information, inventory levels, financial transactions, and other transactional data. The emphasis is on maintaining data integrity, ensuring concurrent access, and supporting transactional **consistency**.

The characteristics of operational databases include:

- 1.Quick Write:** Operational databases prioritize fast data insertion and update operations to ensure real-time transactional processing.
- 2.Quick Read:** These databases are designed for efficient retrieval of specific, up-to-date data needed for transactional operations. They provide fast access to individual records or small subsets of data.

Characteristics – Operational - DW



A **data warehouse**, on the other hand, is a large-scale repository that stores **historical and aggregated data from multiple sources**. It serves as a central location for **data analysis** and reporting, enabling organizations to make informed business decisions. Data warehouses are optimized for **complex queries and analytics rather than transactional processing**.

The characteristics of data warehouses include:

1.One Write: Data warehouses typically follow a batch-oriented approach, where data is extracted, transformed, and loaded (**ETL**) from various operational systems into the warehouse at **regular intervals(not realtime)**. This process involves one-time writes of large volumes of data.

2.Many Read: Once the data is loaded into the data warehouse, it is available for analysis and reporting by **multiple users across the organization**. Data warehouses are optimized for complex queries that involve aggregations, joins, and historical analysis. These queries often involve large datasets and are executed by business analysts or data scientists to gain insights and support decision-making.

(Operational)Database – Data Warehouse

How do you know if your design is suppose to be Operational or DW, Databases and data warehouses are both systems that store data. But they serve very different purposes.

A database stores real-time information about one particular part of your business: its main job is to process the daily transactions that your company makes, e.g., recording which items have been sold. Databases handle a massive volume of simple queries very quickly.

A data warehouse is a system that pulls together data from many different sources within an organization for reporting and analysis. The reports created from complex queries within a data warehouse are used to make business decisions.

A data warehouse stores historical data about your business so that you can analyze and extract insights from it. It does not store current information, nor is it updated in real-time.



(Operational)Database – Data Warehouse – Continued !



Processing Types : OLAP vs OLTP

The most significant difference between databases and data warehouses is how they process data.

Databases use OnLine Transactional Processing (OLTP) to delete, insert, replace quickly, and update large numbers of short online transactions. This type of processing immediately responds to user requests and is used to process a business's day-to-day operations in real-time. For example, if a user wants to reserve a hotel room using an online booking form, the process is executed with OLTP.

Data warehouses use OnLine Analytical Processing (OLAP) to analyze massive volumes of data rapidly. This process gives analysts the power to look at your data from different points of view. For example, even though your database records sales data for every minute of every day, you may want to know the total amount sold each day. To do this, you need to collect and sum the sales data together for each day. OLAP is designed to do this and use it for data warehousing 1000x faster than if you used OLTP to perform the same calculation.



(Operational)Database – Data Warehouse – Continued !



Optimization



A operational database is optimized to **update** (add, modify, or delete) data with maximum speed and efficiency. Response times from databases need to be extremely quick for efficient transaction processing. The most important aspect of a database is that it records the write operation in the system; a company won't be in business very long if its database didn't make a record of every purchase!

Data warehouses are optimized to rapidly **execute a low number of complex read queries** on large multi-dimensional datasets.



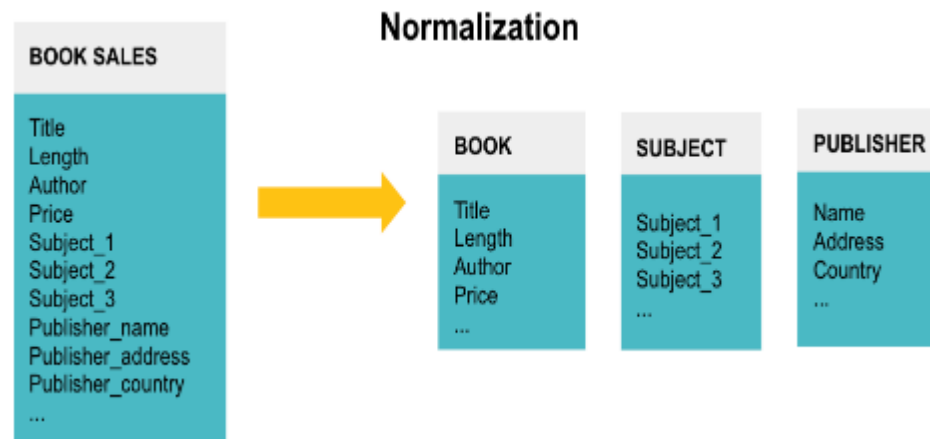
(Operational)Database – Data Warehouse – Continued !



Data Structure

The data in databases are **normalized**. The goal of normalization is to reduce and even eliminate data redundancy, i.e., storing the same piece of data more than once. This reduction of duplicate data leads to increased consistency and, thus, more accurate data, as the database stores it in only one place.

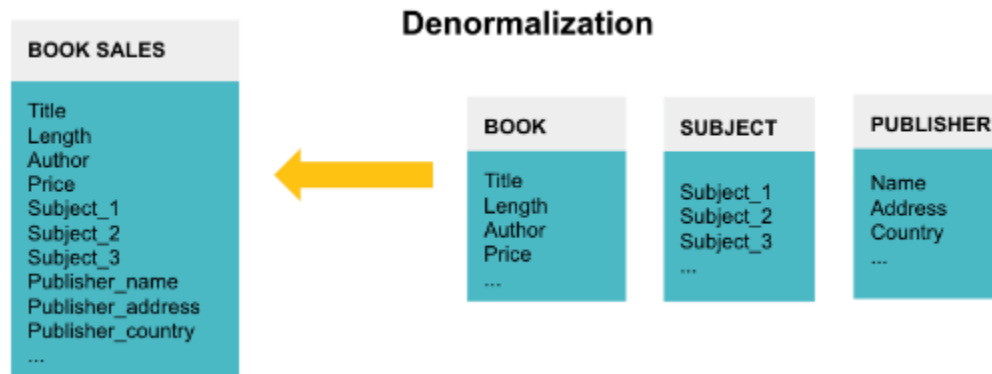
Normalizing data splits it into many different tables. Each table represents a separate entity of the data. For example, a database recording BOOK SALES may have three tables to denote BOOK information, the SUBJECT covered in the book, and the PUBLISHER.



(Operational)Database – Data Warehouse – Continued !

Data Structure

Normalizing data ensures the database takes up minimal disk space and so it is memory efficient. However, it is not query efficient. Querying a normalized database can be slow and cumbersome. Since businesses want to perform complex queries on the data in *their data warehouse*, that data is often denormalized and contains repeated data for easier access.



(Operational)Database – Data Warehouse – Continued !



Data Analysis

Databases usually process transactions, but it is also possible to perform data analysis with them. However, in-depth exploration is challenging for both the user and computer due to the normalized data structure and many tables joins you need to perform. It requires a skilled developer or analyst to create and execute complex queries on a DataBase Management System (DBSM), which takes up a lot of time and computing resources. Moreover, the analysis does not go deep - the best you can get is a one-time static report as databases just give a snapshot of data at a specific time.

Data warehouses are designed to perform complex analytical queries on large multi-dimensional datasets in a straightforward manner. There is no need to learn advanced theory or how to use sophisticated DBMS software. Not only is the analysis simpler to perform, but the results are much more useful; you can dive deep and see how your data changes over time, rather than the snapshot that databases provide.



(Operational)Database – Data Warehouse – Continued !



Data Timeline

Databases process the **day-to-day transactions** for one aspect of the business. Therefore, they typically contain current, rather than historical data about one business process.

Data warehouses are used for analytical purposes and business reporting. Data warehouses typically store **historical data** by integrating copies of transaction data from disparate sources. Data warehouses can also use real-time data feeds for reports that use the most current, integrated



(Operational)Database – Data Warehouse – Continued !



Concurrent Users

Database support thousands of concurrent users because they are updated in real-time to reflect the business's transactions. Thus, many users need to interact with the database simultaneously without affecting its performance.

However, only one user can modify a piece of data at a time - it would be disastrous if two users overwrite the same information in different ways at the same time!

In contrast, data warehouses support a limited number of concurrent users. A data warehouse is separated from front-end applications, and using it involves writing and executing queries. These queries are computationally expensive, and so only a small number of people can use the system simultaneously.



(Operational)Database – Data Warehouse – Continued !



Database – Use Cases

Databases process the day-to-day transactions in an organization. Some examples of database applications include:

- An ecommerce website creating an order for a product it has sold.
- An airline using an online booking system.
- A hospital registering a patient.
- A bank adding an ATM withdrawal transaction to an account.



(Operational)Database – Data Warehouse – Continued !



Datawarehouse – Use Cases

Data warehouses provide high-level reporting and analysis that empower businesses to make more informed business.

Use cases include:

- Segmenting customers into different groups based on their past purchases to provide them with more tailored content.
- Predicting customer churn using the last ten years of sales data
- Creating demand and sales forecasts to decide which areas to focus on next quarter.



Please also refer to references slide at the end of this presentation for credits and citations for the contents that are not exactly copied but taken as a knowledge or reference

OLAP and OLTP Systems



Datawarehouse – Use Cases

OLTP is an online data modification system, while **OLAP** is an online historical multidimensional data retrieval system, which retrieves the data for analysis that can help in decision making. Which one to use depends upon the user's requirement both works for different purpose.

OLTP is an **Online Transaction Processing system**. The primary focus of OLTP system is to record the current **Update, Insertion and Deletion** while transaction. The OLTP queries are **simpler** and **short** and hence require **less time in processing** and also require **less space**.

OLTP database gets **updated frequently**. It may happen that a transaction in OLTP fails in middle, which may affect **data integrity**. So, it has to take special care of data integrity. OLTP database has **normalized tables** (3NF).

The best example for OLTP system is an **ATM**, in which using quick transactions we change the status of our account. OLTP system becomes the source of data for OLAP.

OLAP is an **Online Analytical Processing system**. OLAP database stores historical data that has been inputted by OLTP. It allows a user to view different summaries of multi-dimensional data. Using OLAP, you can extract information from an extensive database and analyze it for decision making.

OLAP also allows a user to execute **complex queries** to extract multidimensional data. In OLTP, even if the transaction fails in middle, it will not harm data integrity as the user uses OLAP system to retrieve data from a large database to analyze. Simply the user can fire the query again and extract the data for analysis.

The transaction in OLAP are **long** and hence take comparatively **more time** for processing and require large space. The transactions in OLAP are **less frequent** as compared to OLTP. Even the tables in OLAP database may not be normalized. The example for OLAP is to view a financial report, or budgeting, marketing management, sales report, etc.



Credits and Citations

- Star Schema and Snowflake schema in Datawarehouse

<https://www.guru99.com/star-snowflake-data-warehousing.html>

- Database vs Data Warehouse

<https://panoply.io/data-warehouse-guide/the-difference-between-a-database-and-a-data-warehouse/>

- OLAP and OLTP

<https://techdifferences.com/difference-between-oltp-and-olap.html>